



12. Autumn Coin

Team Olomoucké srnky, Gymnázium, Olomouc-Hejčín

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The motion of a coin falling to the bottom of a tank filled with liquid can be remarkably similar to the fluttering and tumbling of a falling autumn leaf. Investigate how the motion of the coin depends on relevant parameters.



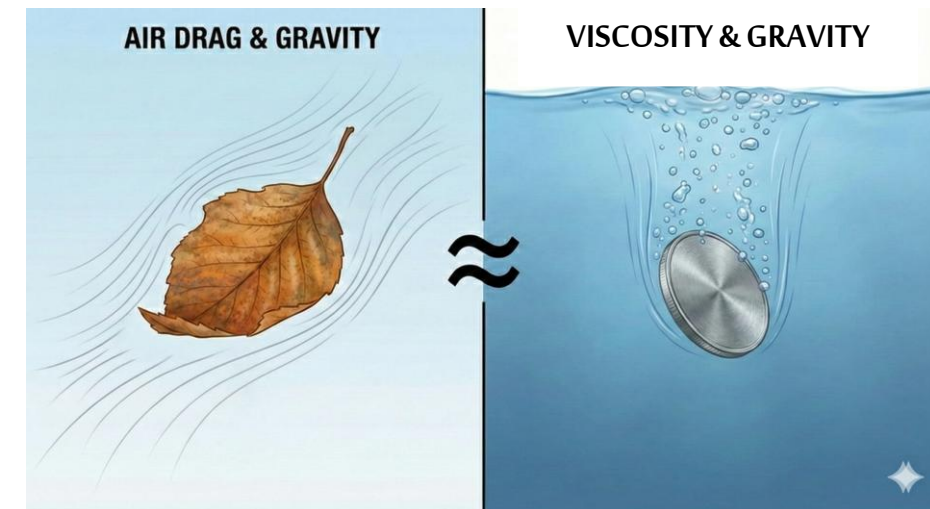
RESEARCH QUESTIONS:

1. How can the descent of the coin be slowed down to maximize the fall time?
2. How can we make the movement of the coin as stable as possible.
3. How to determine the transition between the two types of motion of the coin – tumbling/fluttering and what is the „between“.



WHY SHOULD A HEAVY COIN BEHAVE LIKE A LEAF?

- water is ≈ 800 x denser than air
 - higher fluid viscosity
 - higher mass
- they balance each other out



Generated via Google Gemini AI.



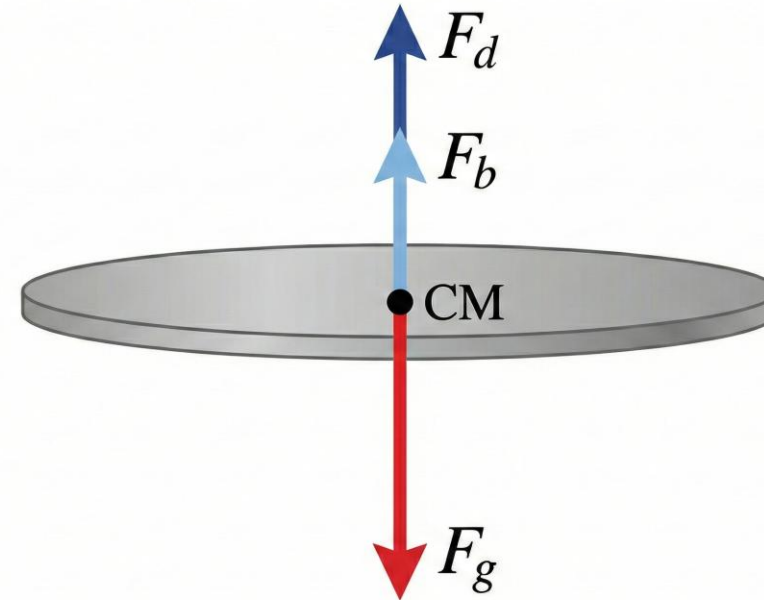
1. IDEAL VERTICAL FALL

- ▣ Symmetric flow → forces balanced at center of mass
- ▣ Reality: unstable equilibrium

$$\sum F_x = 0, \tau = 0$$

F_x = horizontal forces [N]

τ = torque [N·M]



Generated via Google Gemini AI.



2. PRESSURE GRADIENT

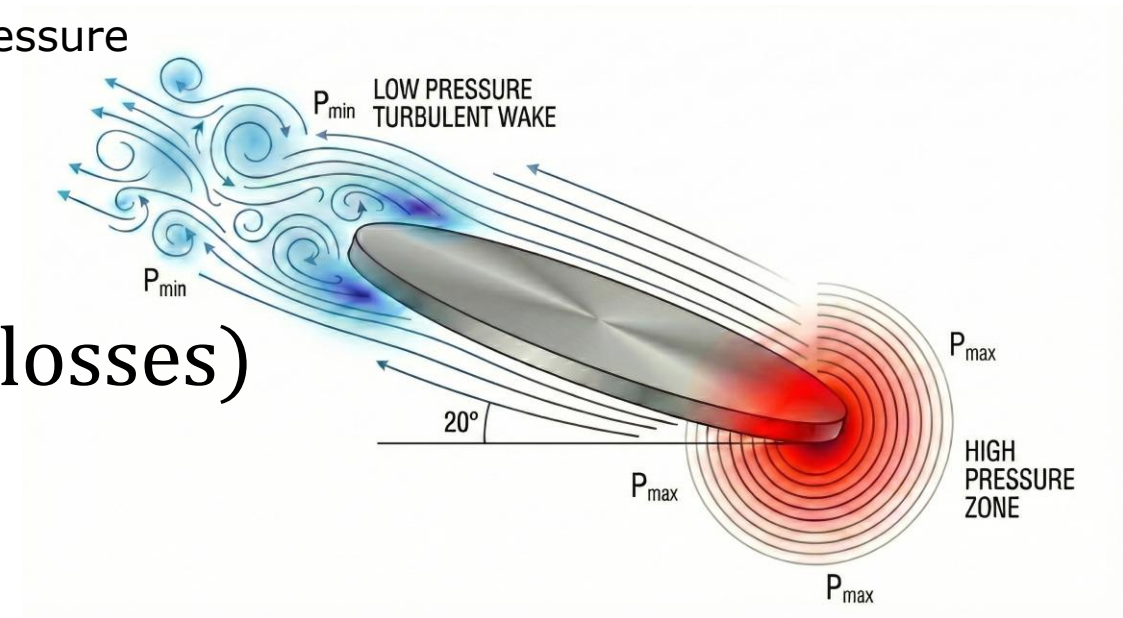
- Small tilt creates **angle of attack**
- Bernoulli principle: Velocity drops → pressure peaks

$$p + \frac{1}{2}\rho v^2 \approx \text{constant (no losses)}$$

p = pressure [Pa]

ρ = fluid density [kg/m^3]

v = fluid velocity



Generated via Google Gemini AI.



3. TORQUE GENERATION

Center of pressure shifts forward

→ Lever arm created

Offset (r): distance between CP and CM

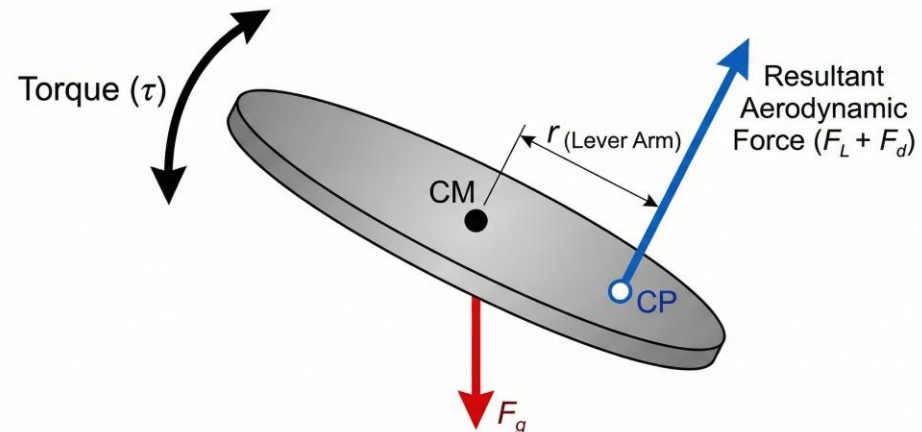
Torque initiates tumbling

$$\tau = r \times F_r$$

F_r = resultant aerodynamic force [N]

τ = torque [N·M]

r = lever arm [m]



Generated via Google Gemini AI.



4. DYNAMICS OF FALLING BODIES

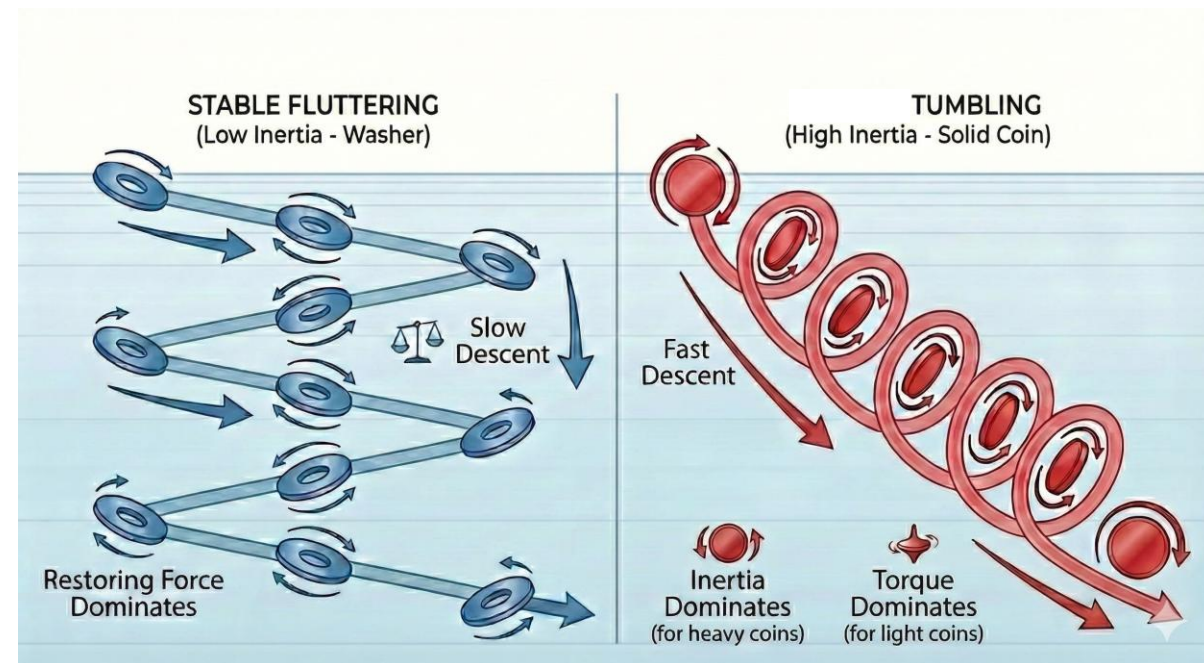
- Restoring torque (τ) vs. Angular inertia
- Tumbling – periodic
- Fluttering – periodic
- Unpredictable alternation between tumbling and fluttering -> chaotic

$$I = \frac{1}{4}mr^2 + \frac{1}{12}mh^2$$

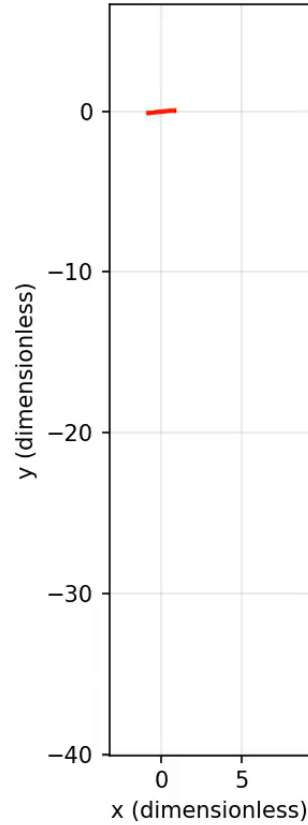
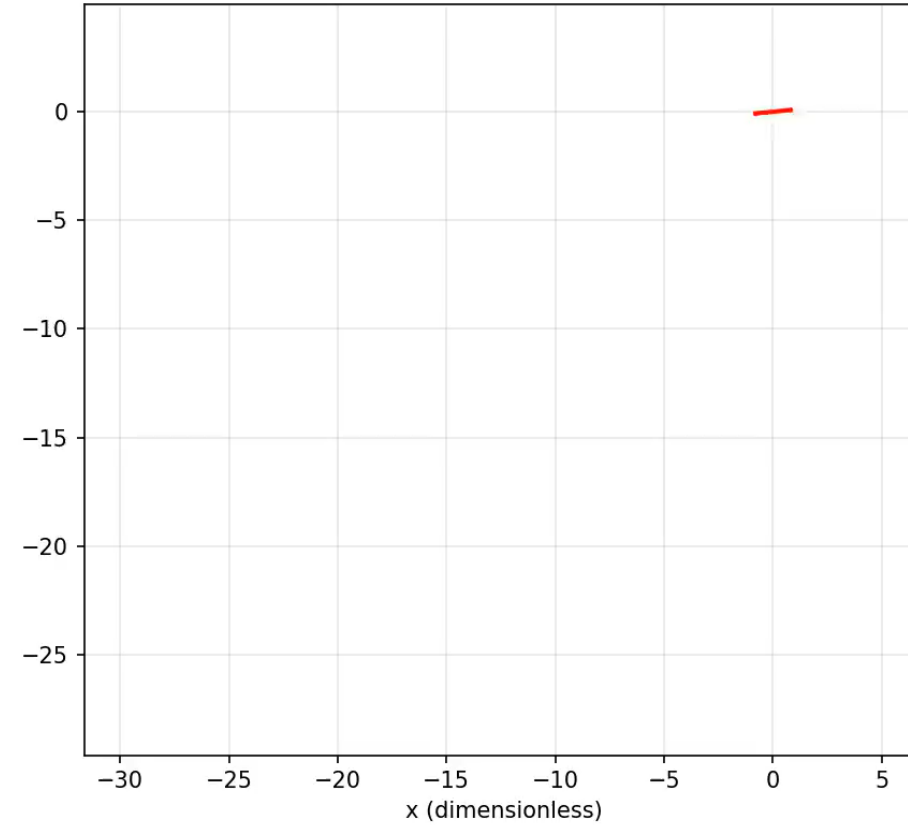
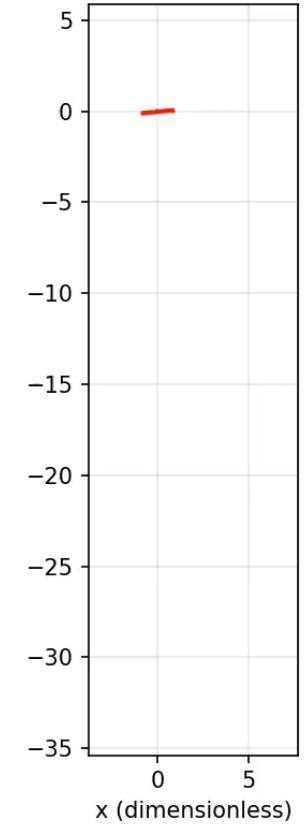
I = moment of inertia [$\text{kg} \cdot \text{m}^2$]

m = mass of the coin [kg]

r = radius of the coin [m]



Generated via Google Gemini AI.

Falling Coin: Fluttering, Tumbling, Periodic Mixture**Fluttering ($I^* = 1.1$)****Tumbling ($I^* = 1.4$)****Periodic Mixture ($I^* = 1.6$)**

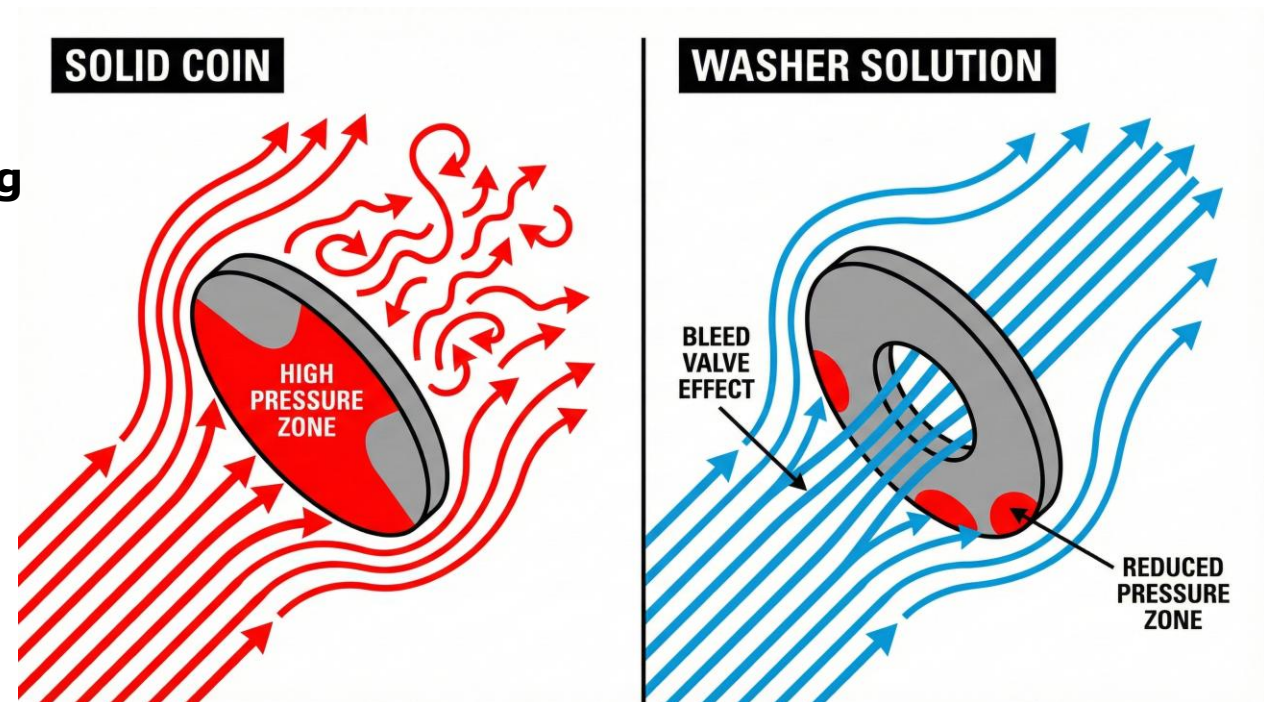


5. THE WASHER SOLUTION

- Center mass removed ($m \downarrow$)
- > inertia reduction
- > easier to dampen the rotation
- Bleed valve effect -> **stable fluttering**

$$I_{washer} < I_{solid}$$

$$\Delta p \approx 0 \rightarrow \tau \approx 0$$



Generated via Google Gemini AI.



6. Important quantities

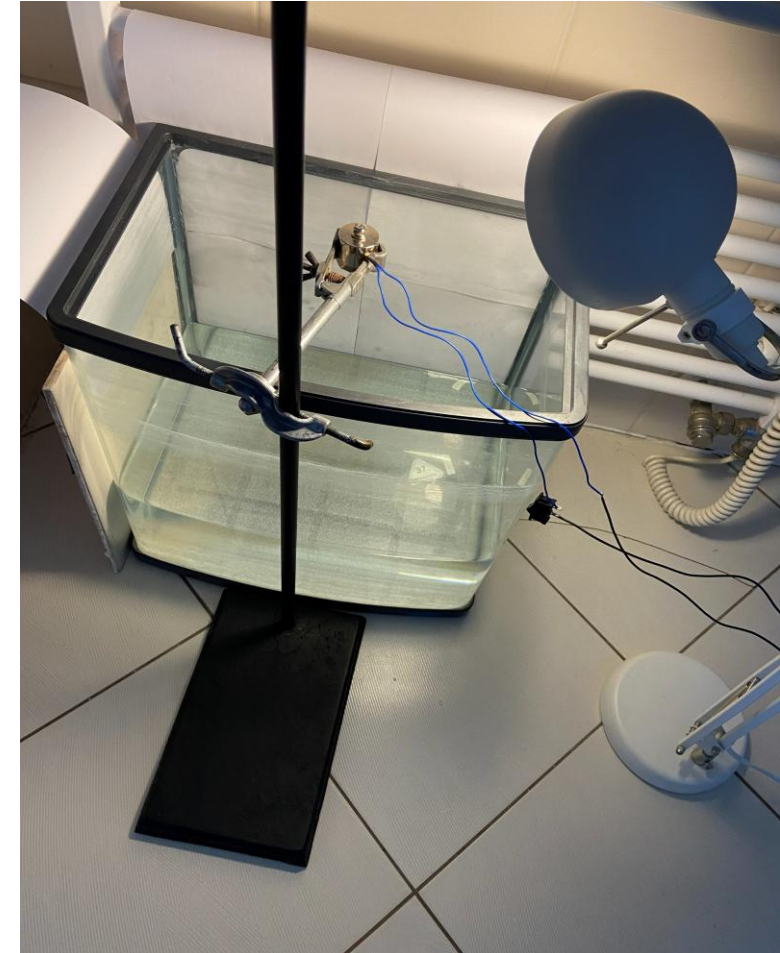
- **Reynolds number** – it represents the ratio of inertial forces to viscous forces
- **Dimensionless moment of inertia** – ratio comparing the coin's own resistance to rotation (I) against the resistance of the surrounding fluid ($\rho \cdot D^5$)

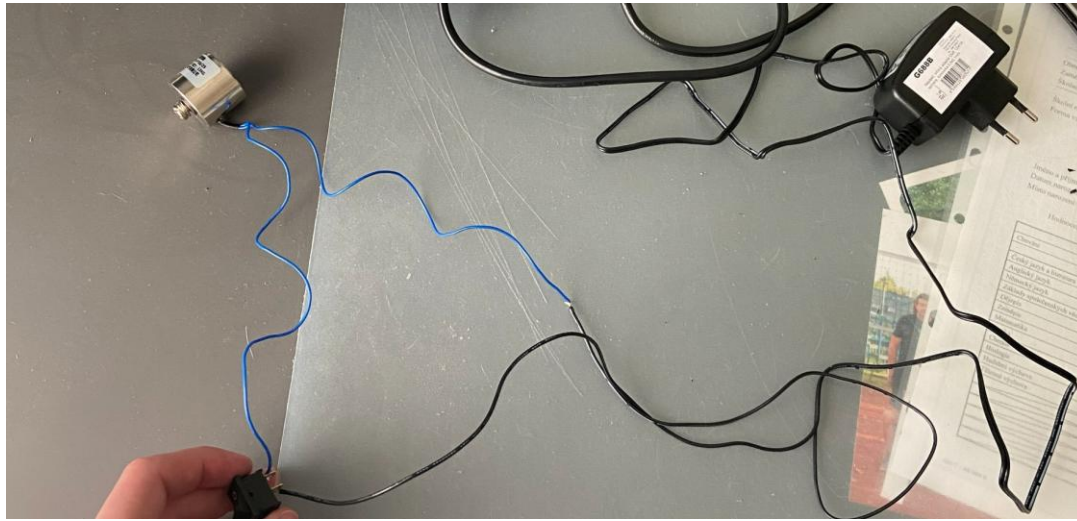
$$Re = \frac{v \cdot D}{\nu} \qquad I^* = \frac{I}{\rho \cdot D^5}$$

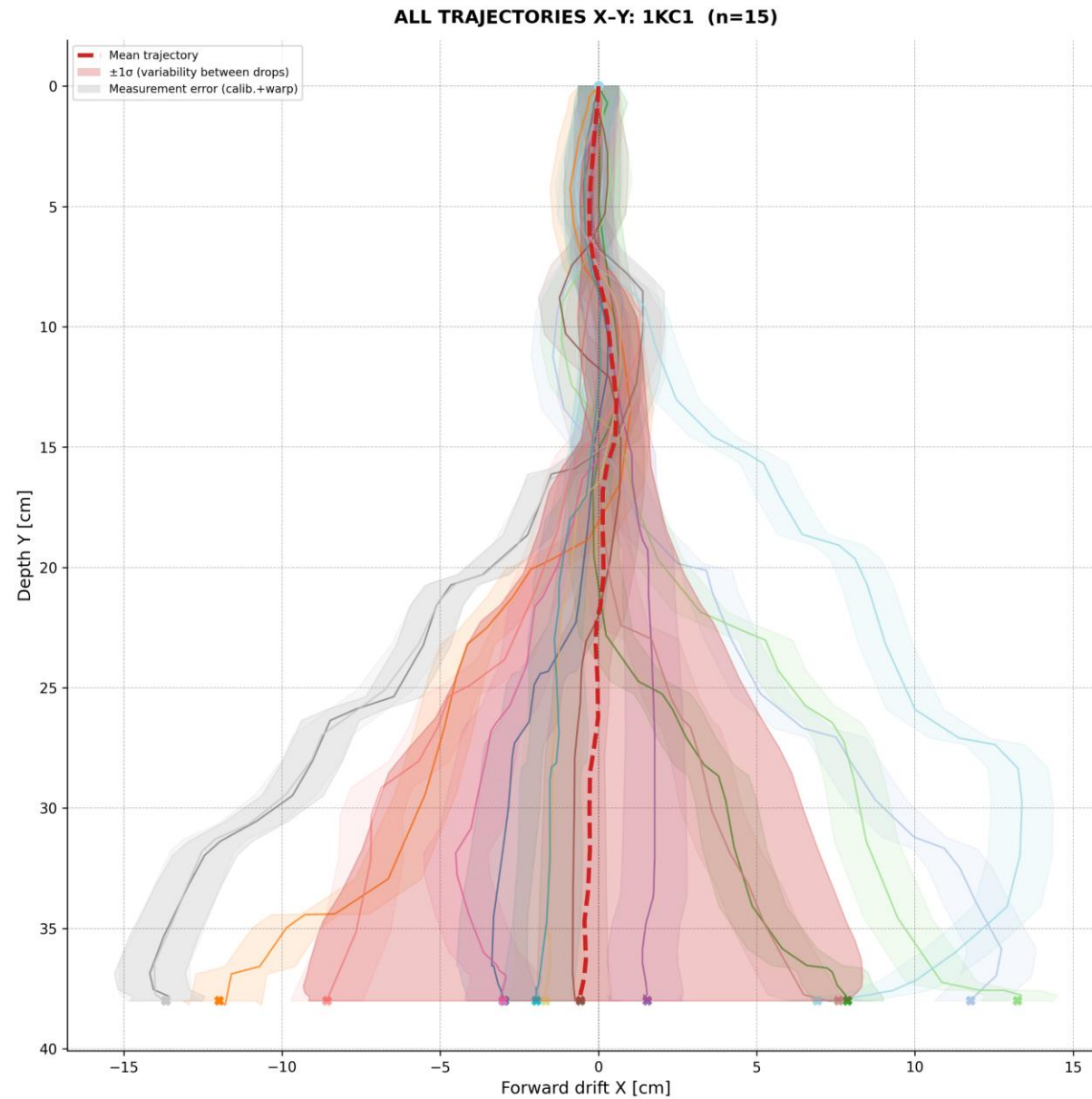


EXPERIMENT SETUP

- water tank (38 x 32 x 38 cm)
- electromagnet
- CZK coins (1, 5, 10, 20, 50 CZK)
- Danish coins
- iPhone 15 Pro camera, slow-motion (240 fps),
Iphone 11 camera, slow-motion (240 fps)
- Tracker* video analysis, python



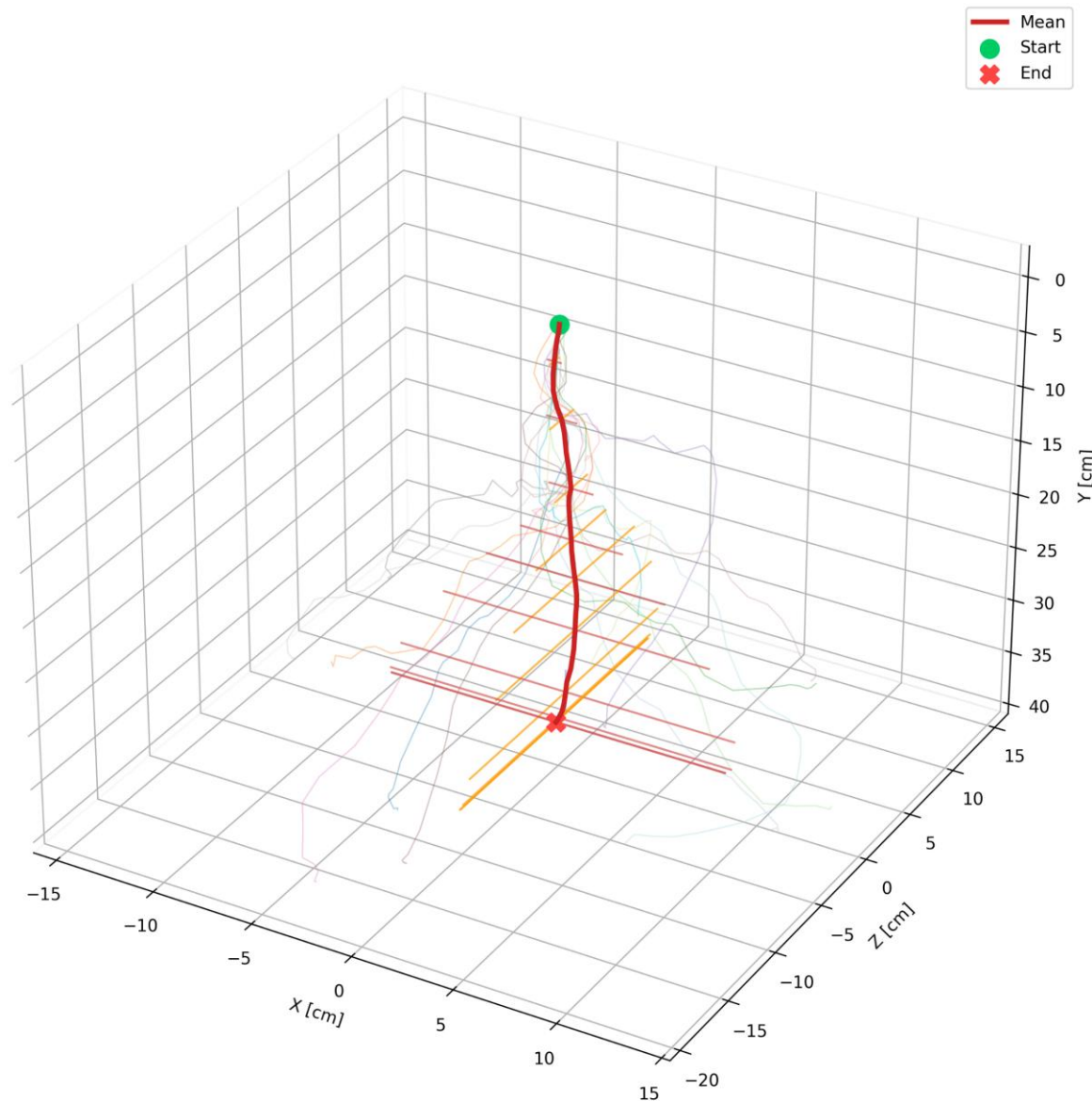




1 CZK

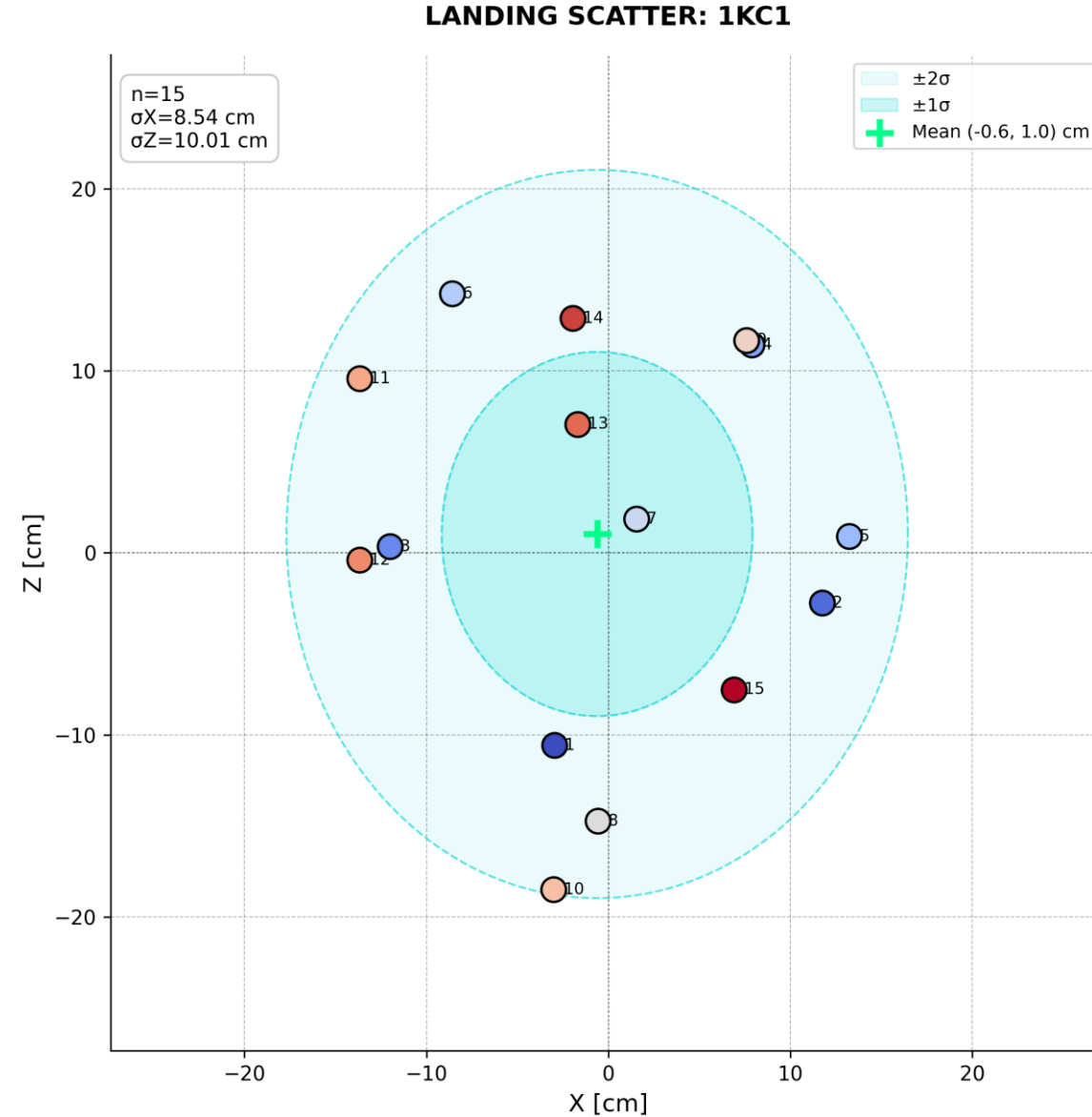
$m = 3.6 \text{ g}$, $d = 20 \text{ mm}$, $n = 15$

3D TRAJECTORIES: 1KC1 (n=15)



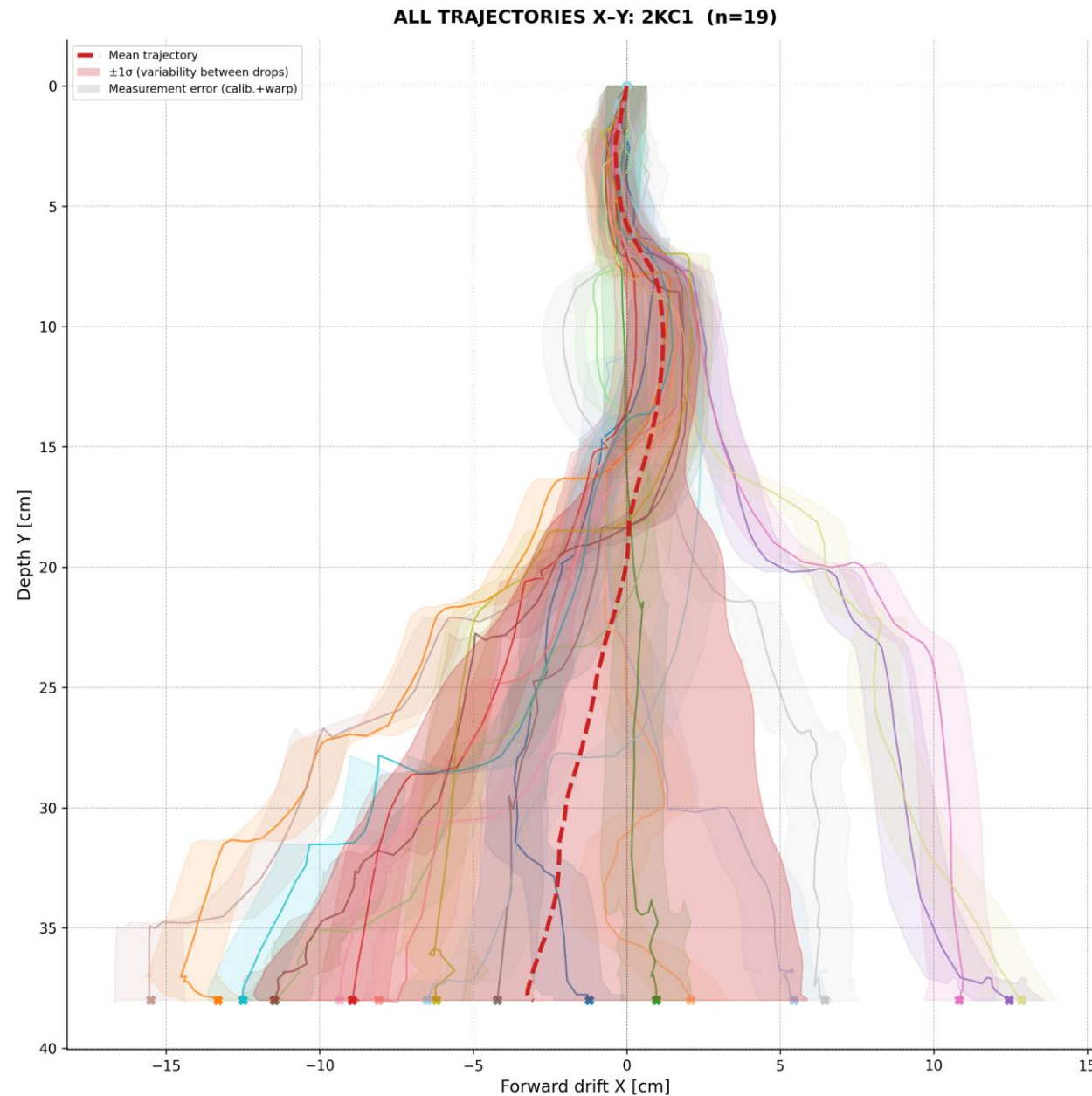
1 CZK

$m = 3.6 \text{ g}$, $d = 20 \text{ mm}$, $n = 15$



1 CZK

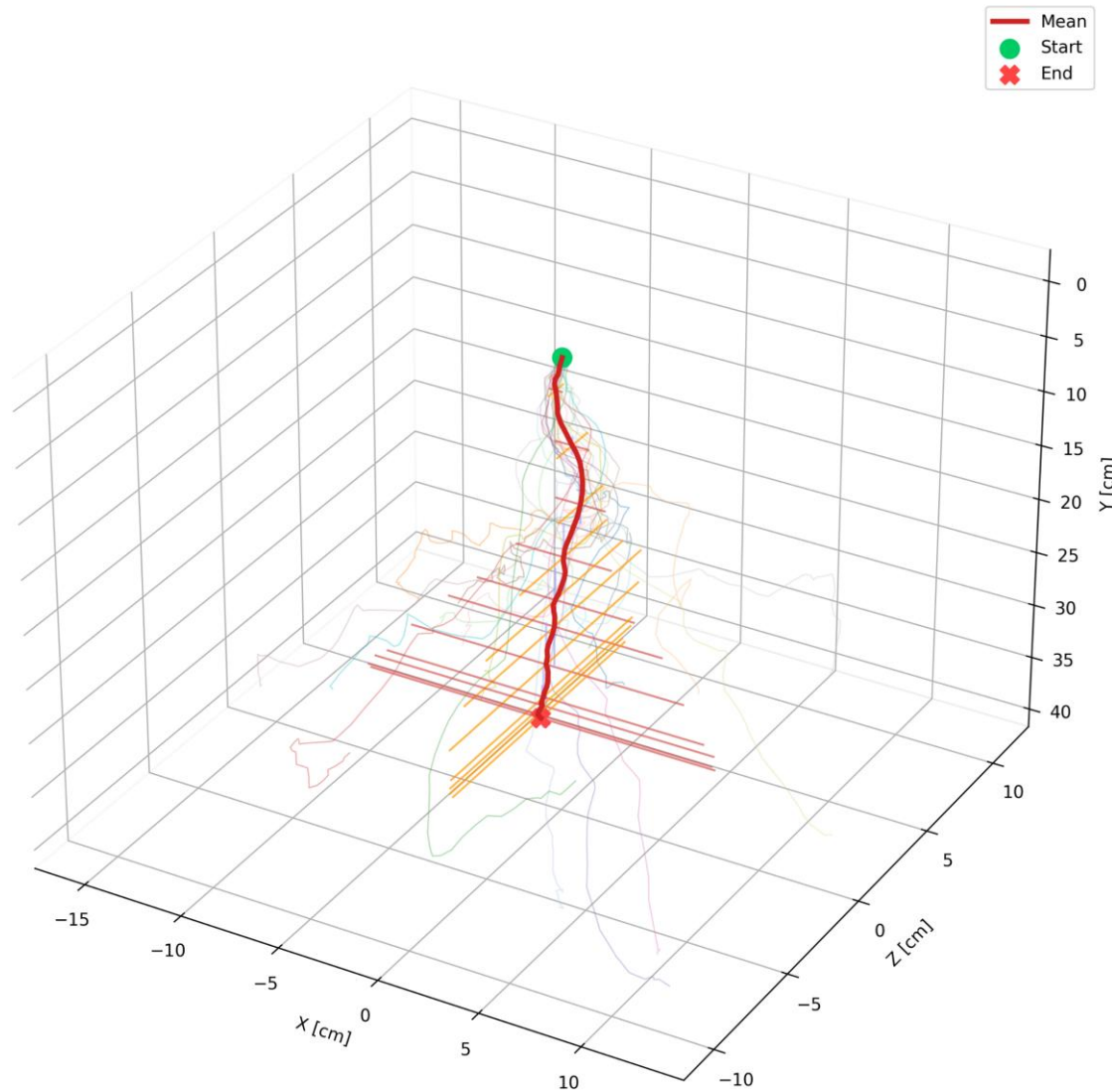
$m = 3.6$ g, $d = 20$ mm



2 CZK

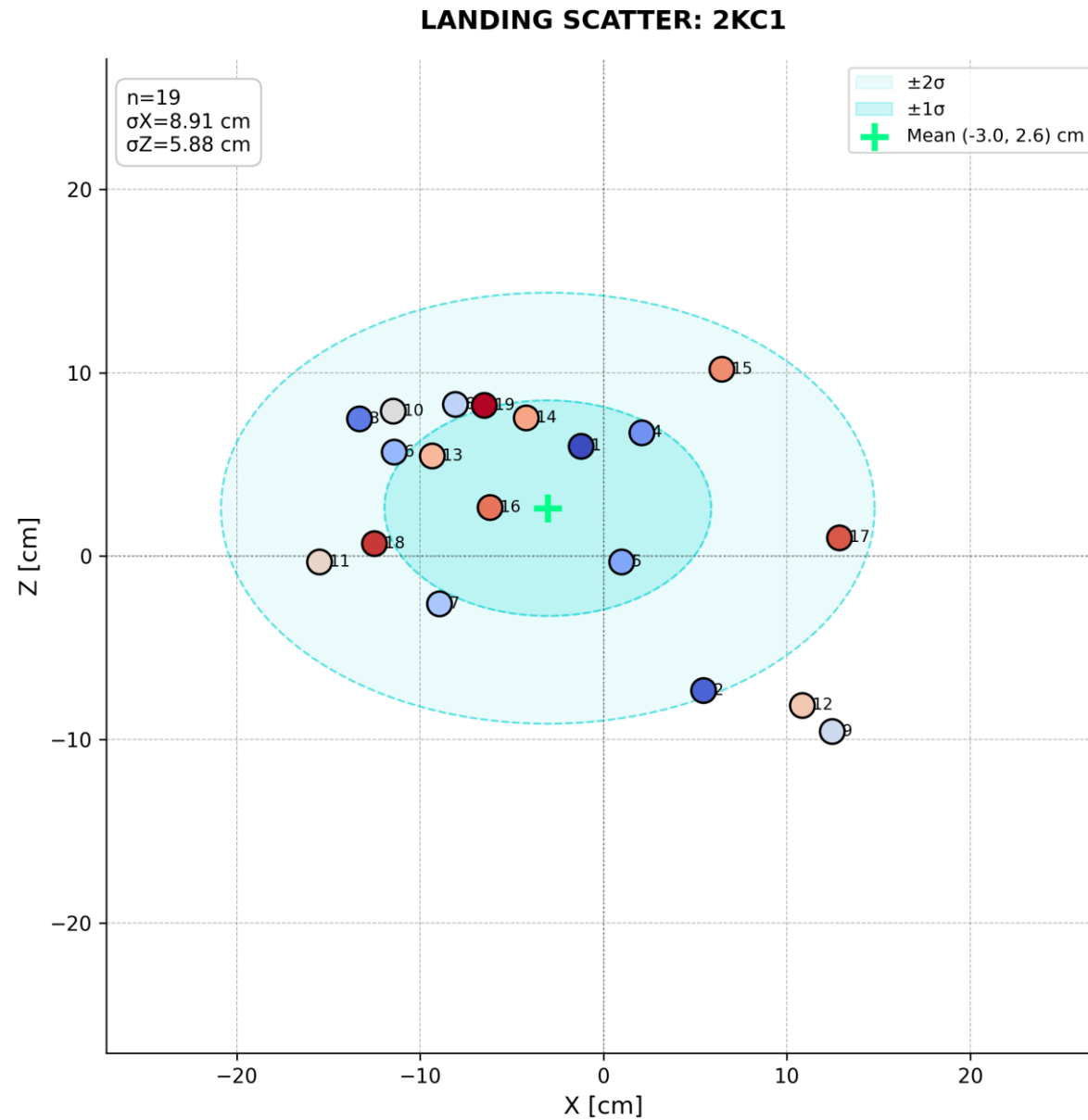
$m = 3.7 \text{ g}$, $d = 21,5 \text{ mm}$ $n = 19$

3D TRAJECTORIES: 2KC1 (n=19)



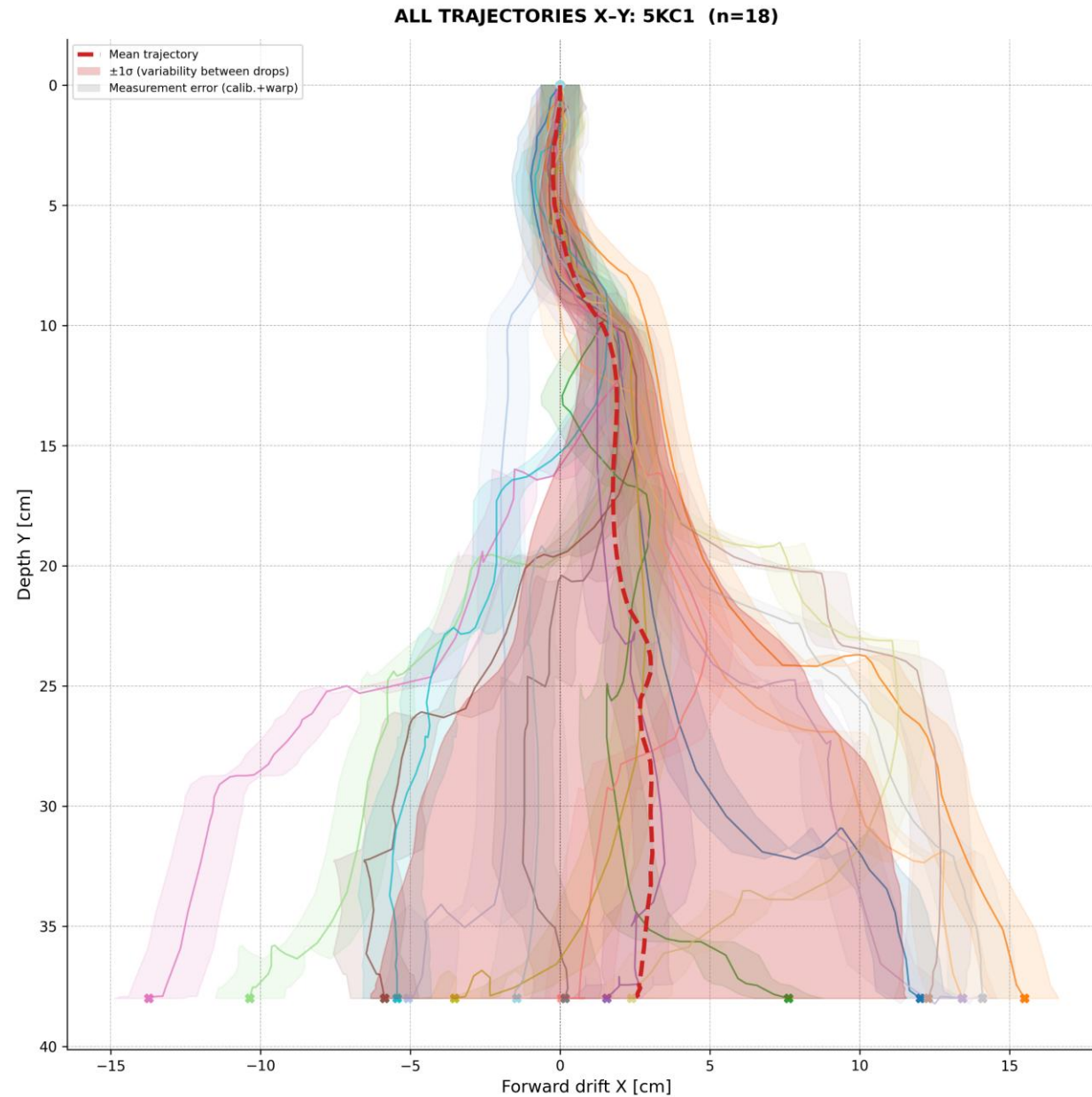
2 CZK

$m = 3.7 \text{ g}$, $d = 21,5 \text{ mm}$, $n = 19$



2 CZK

$m = 3.7$ g, $d = 21,5$ mm, $n = 19$

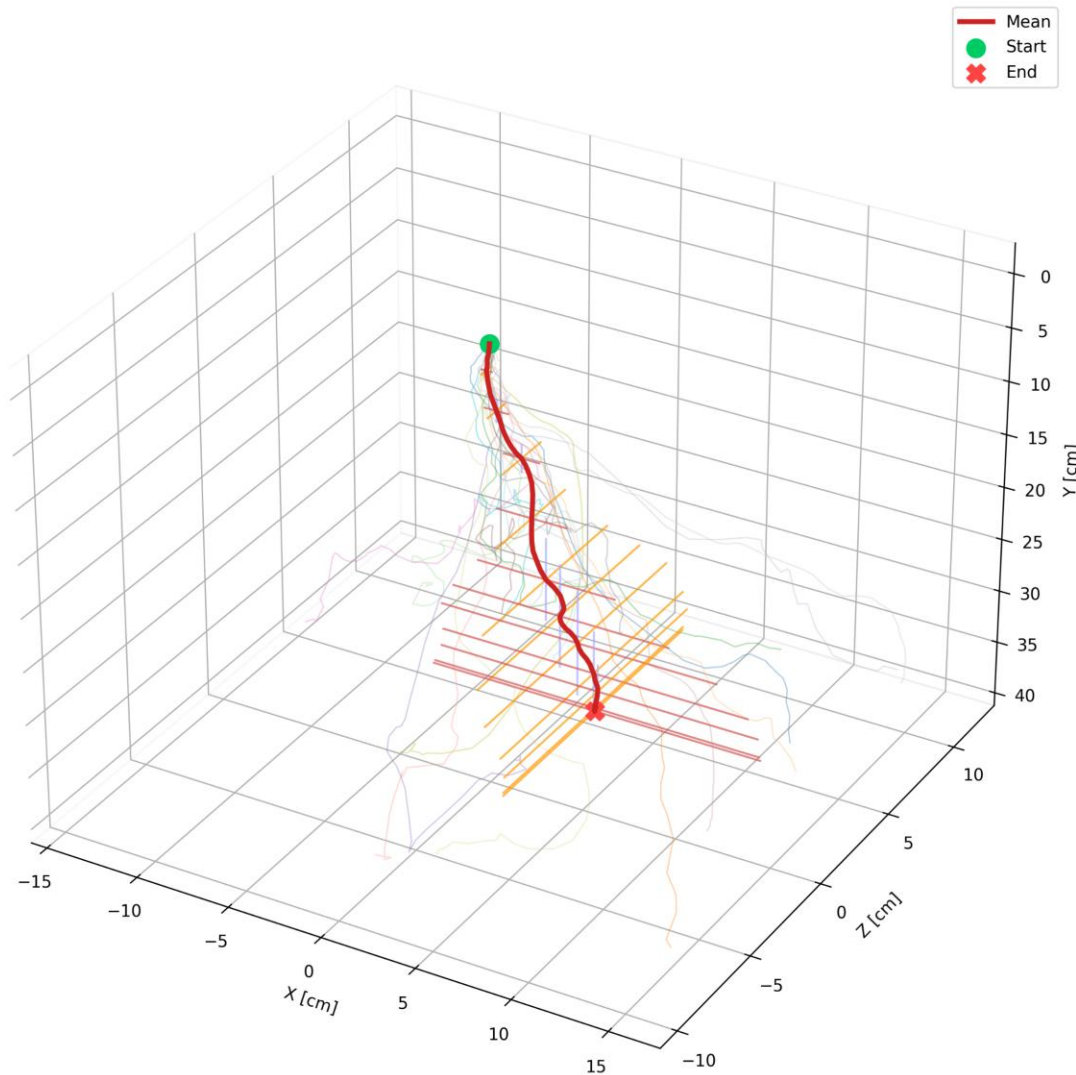


5 CZK

$m = 4,8 \text{ g}$, $d = 23 \text{ mm}$, $n = 18$

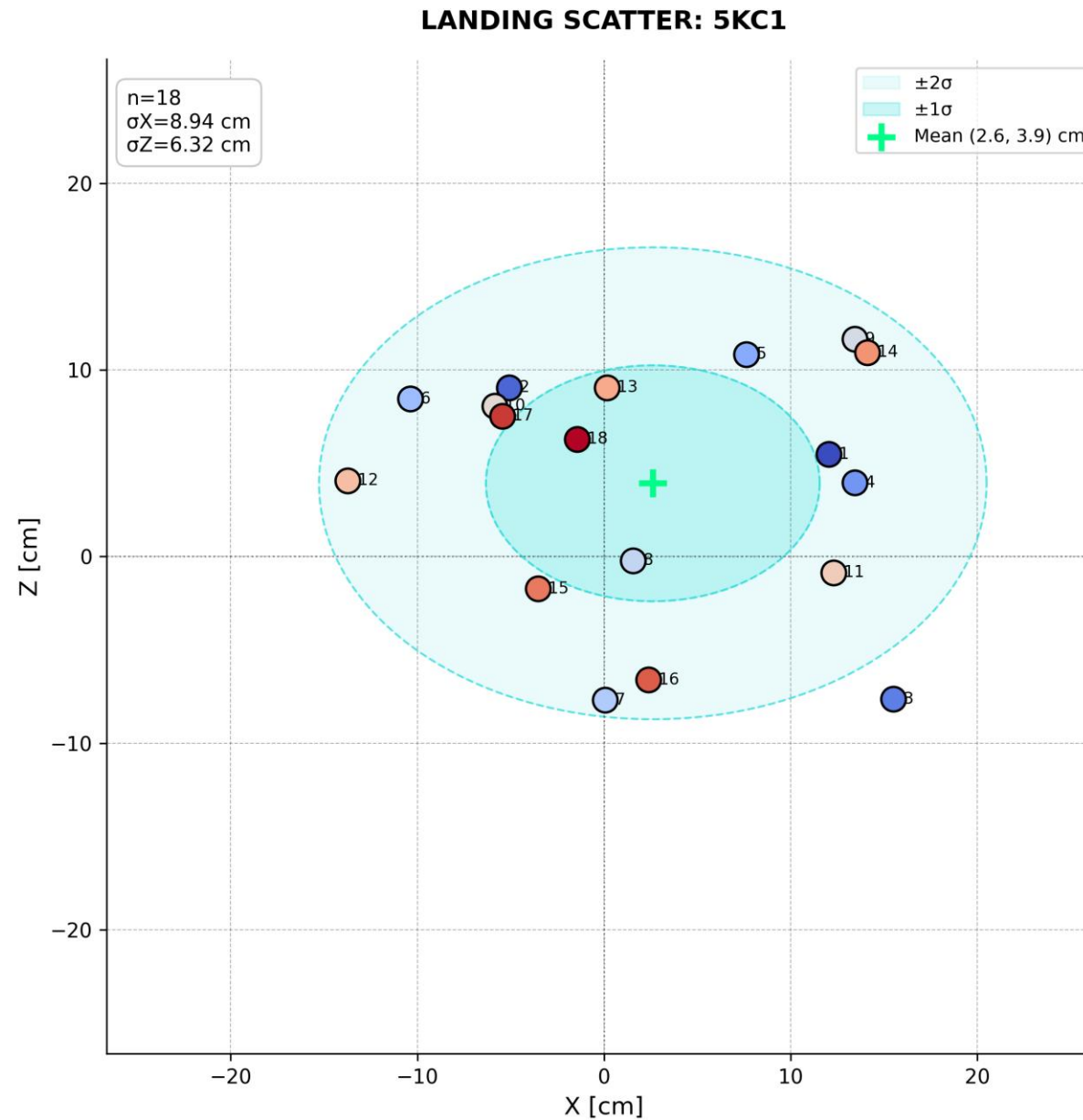


3D TRAJECTORIES: 5KC1 (n=18)



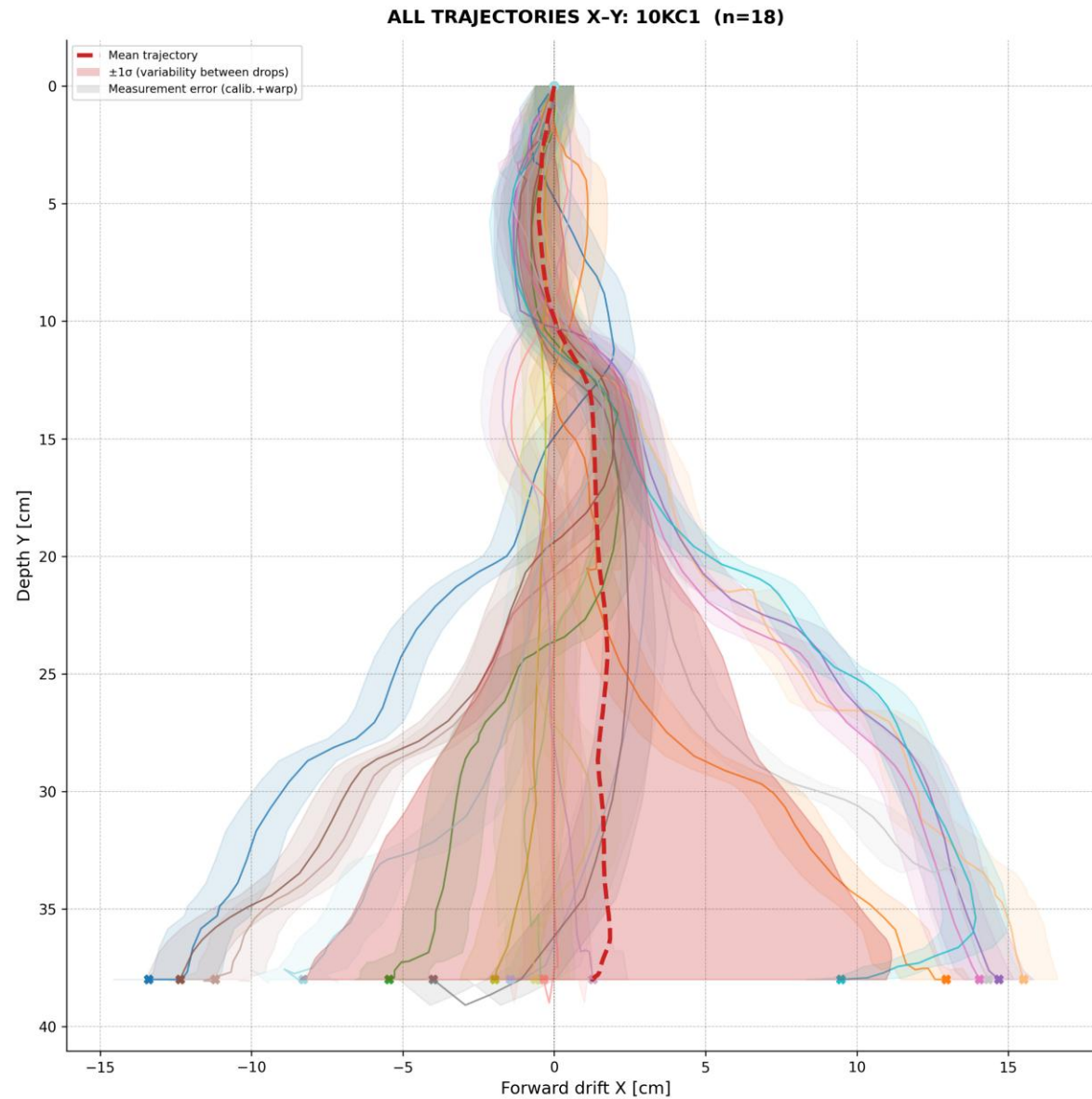
5 CZK

$m = 4,8 \text{ g}$, $d = 23 \text{ mm}$, $n = 18$



5 CZK

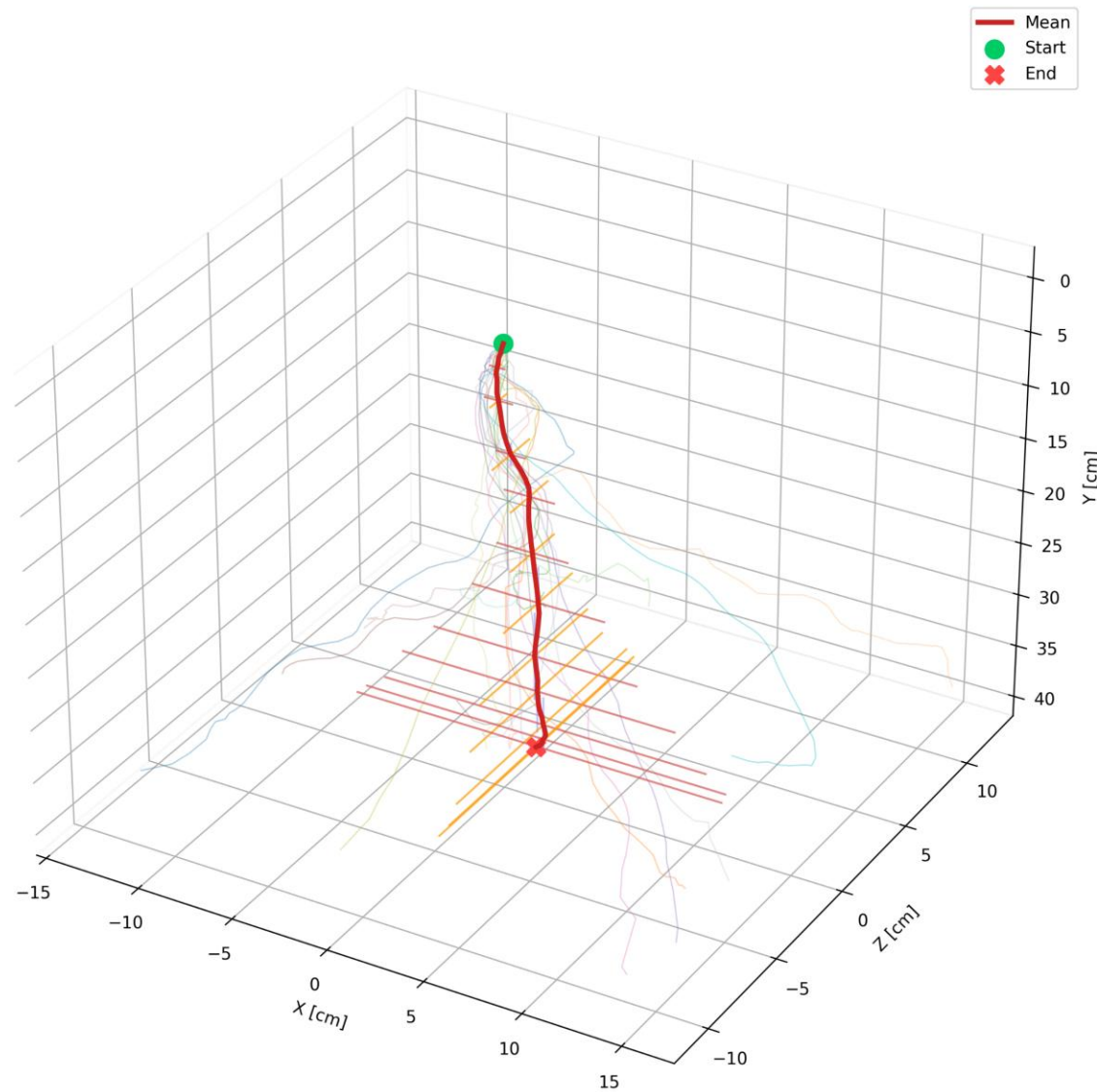
$m = 4,8$ g, $d = 23$ mm $n = 18$



10 CZK

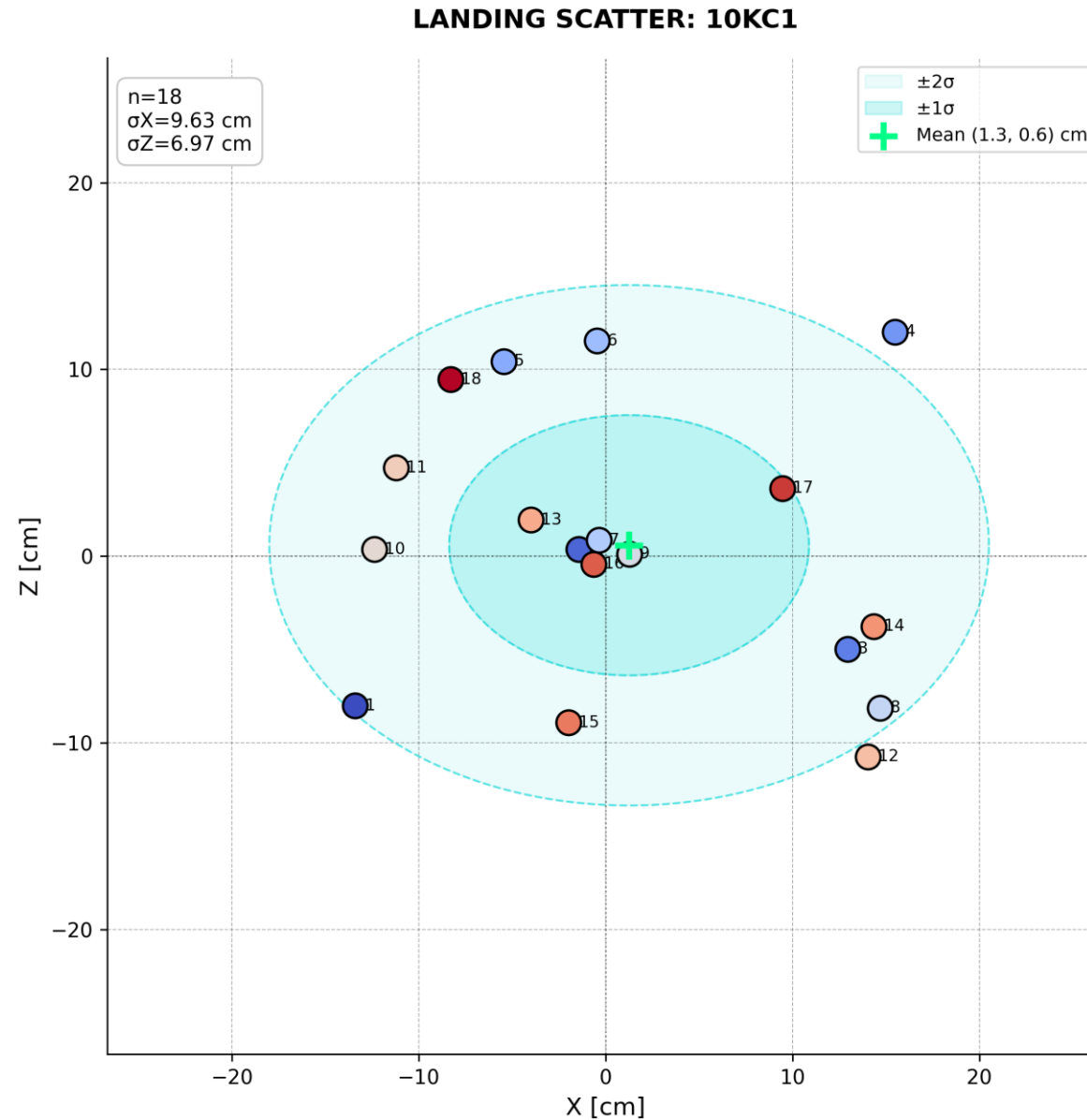
$m = 7.62 \text{ g}$, $d = 24.5 \text{ mm}$ $n = 18$

3D TRAJECTORIES: 10KC1 (n=18)



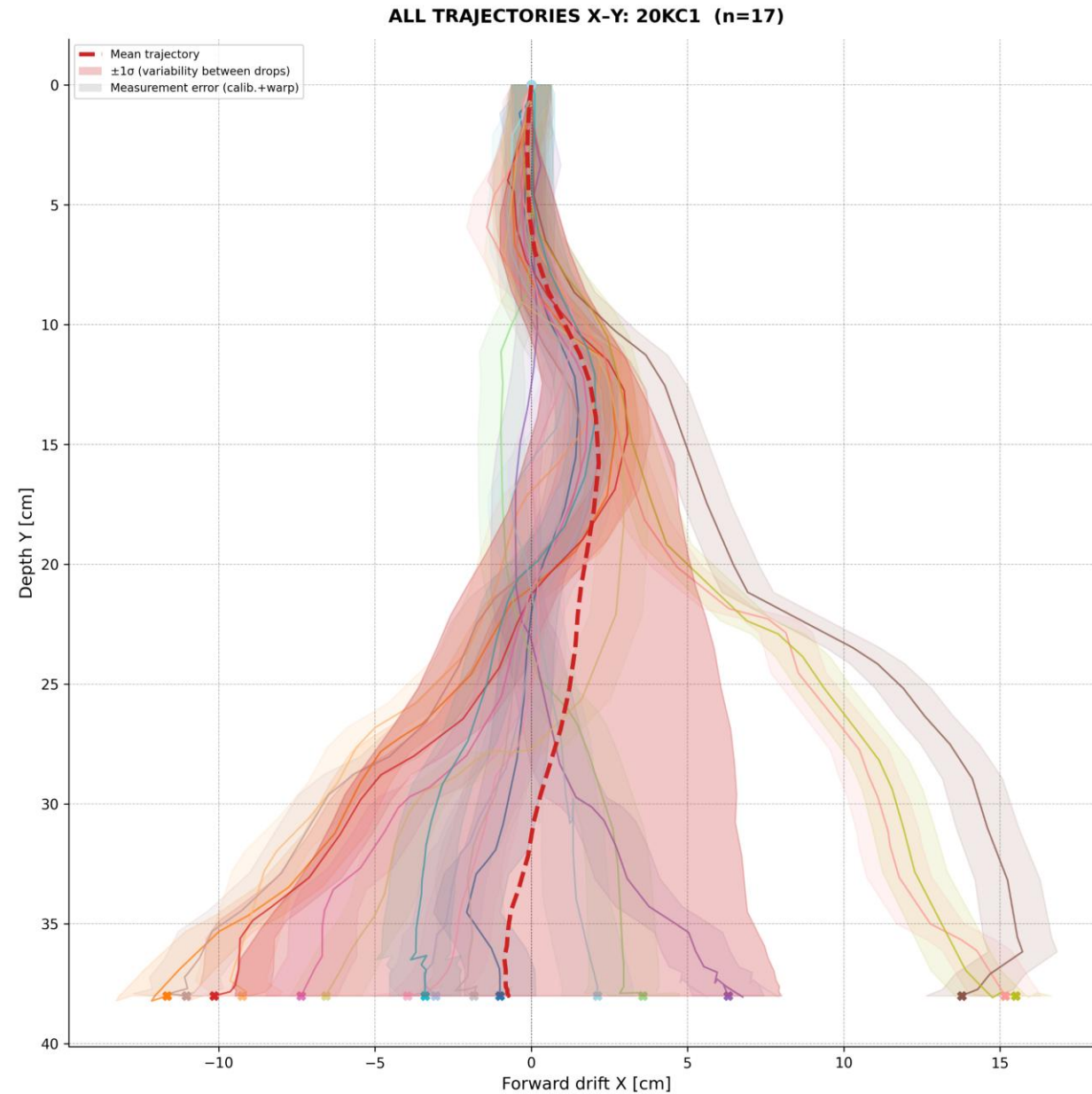
10 CZK

$m = 7.62 \text{ g}$, $d = 24.5 \text{ mm}$ $n = 18$



10 CZK

m = 7.62 g, d = 24.5 mm n = 18

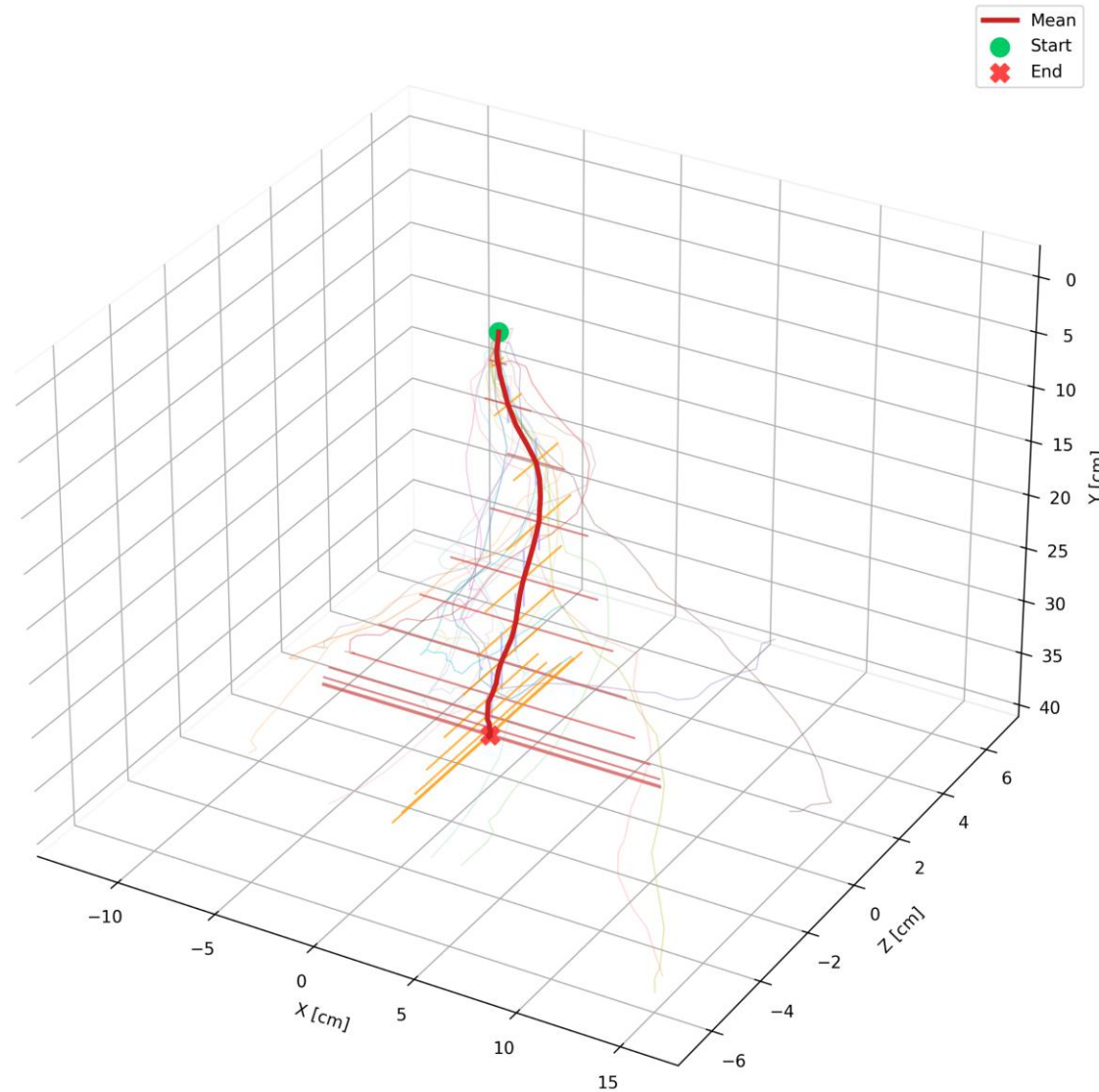


20 CZK

m = 8.42 g, d = 26 mm n = 17

Plot updated after the tournament with corrected data

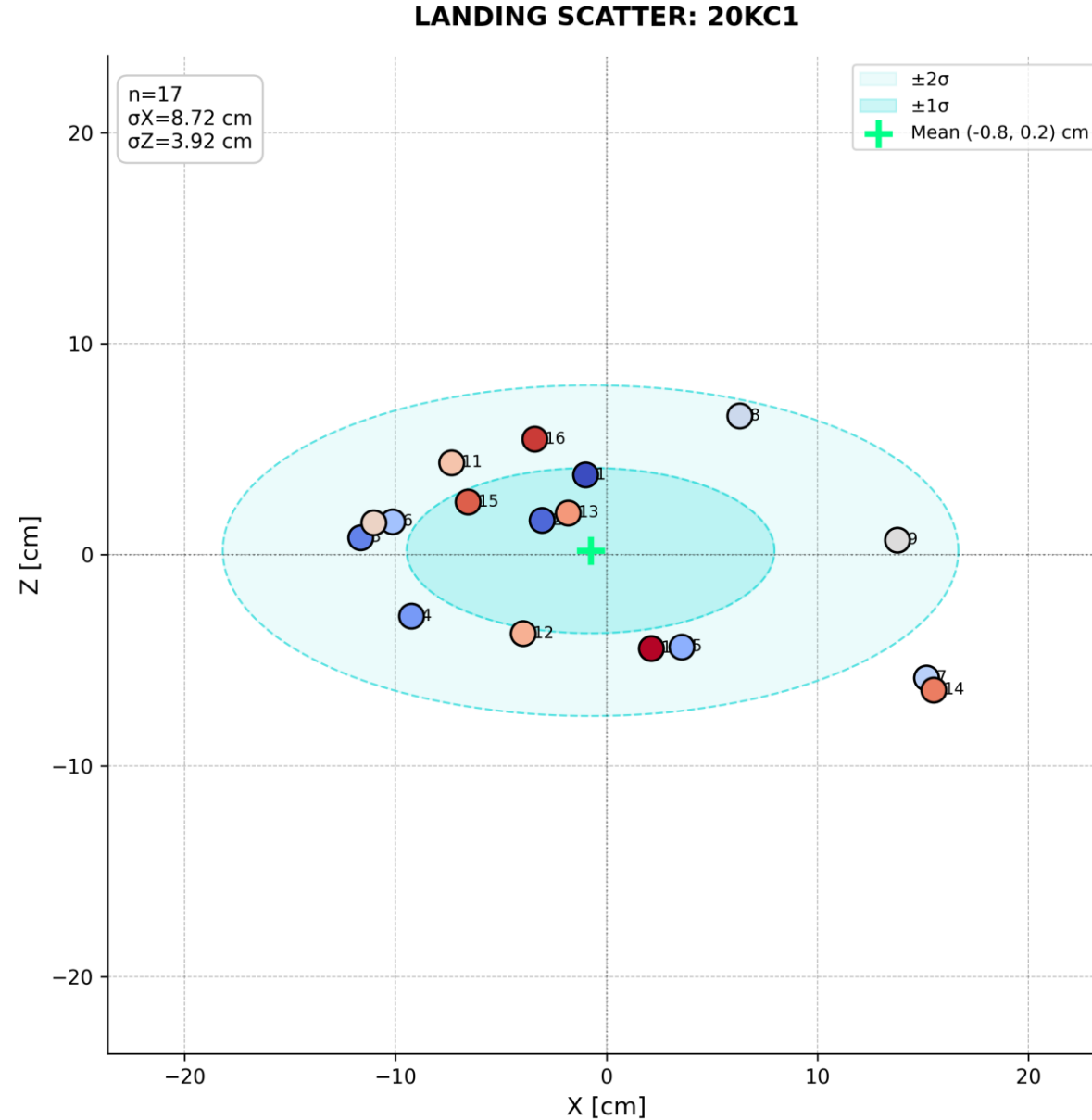
3D TRAJECTORIES: 20KC1 (n=17)



20 CZK

$m = 8.42 \text{ g}$, $d = 26 \text{ mm}$ $n = 17$

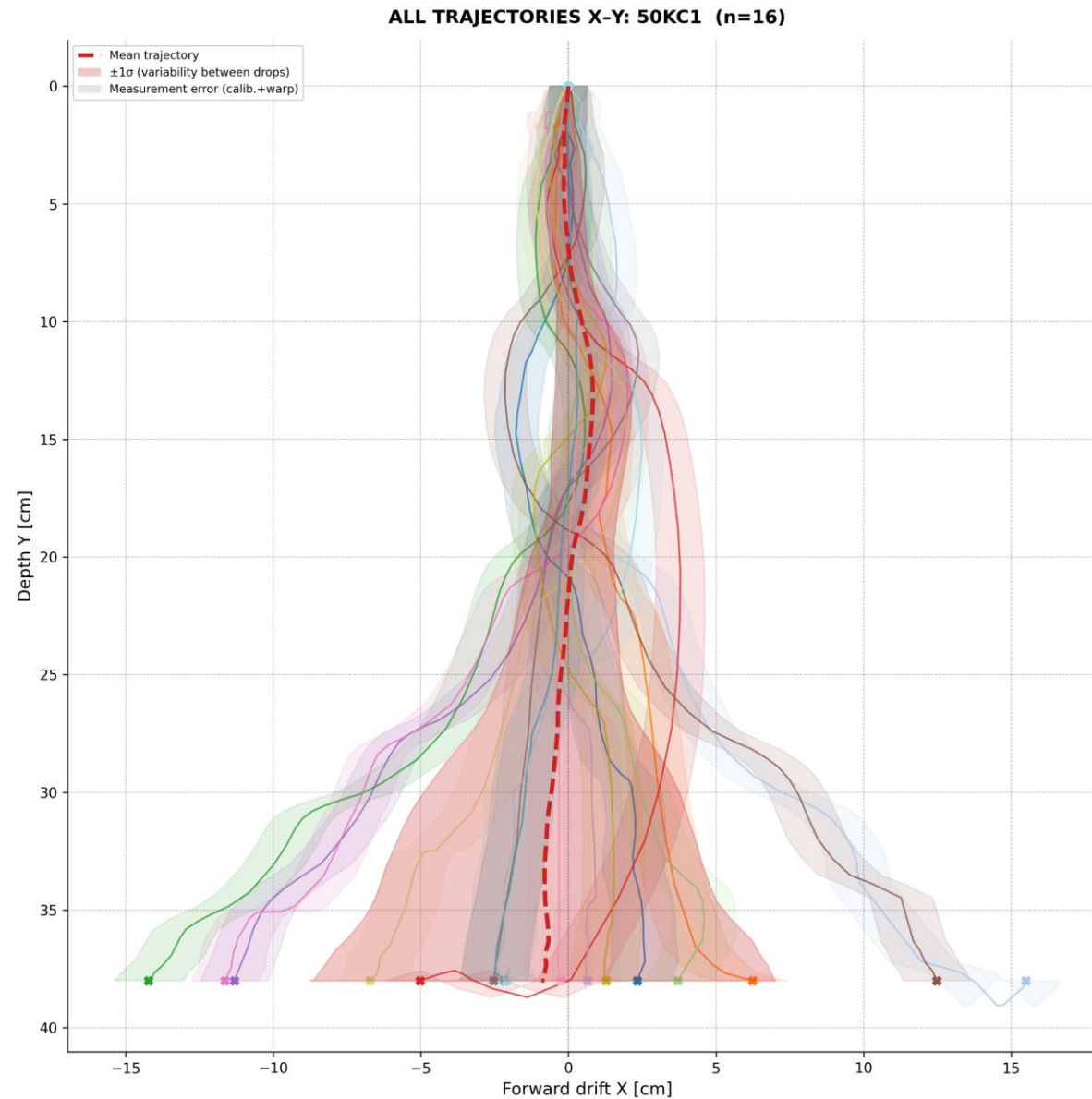
Plot updated after the tournament with corrected data



20 CZK

$m = 8.42 \text{ g}$, $d = 26 \text{ mm}$ $n = 17$

Plot updated after the tournament with corrected data

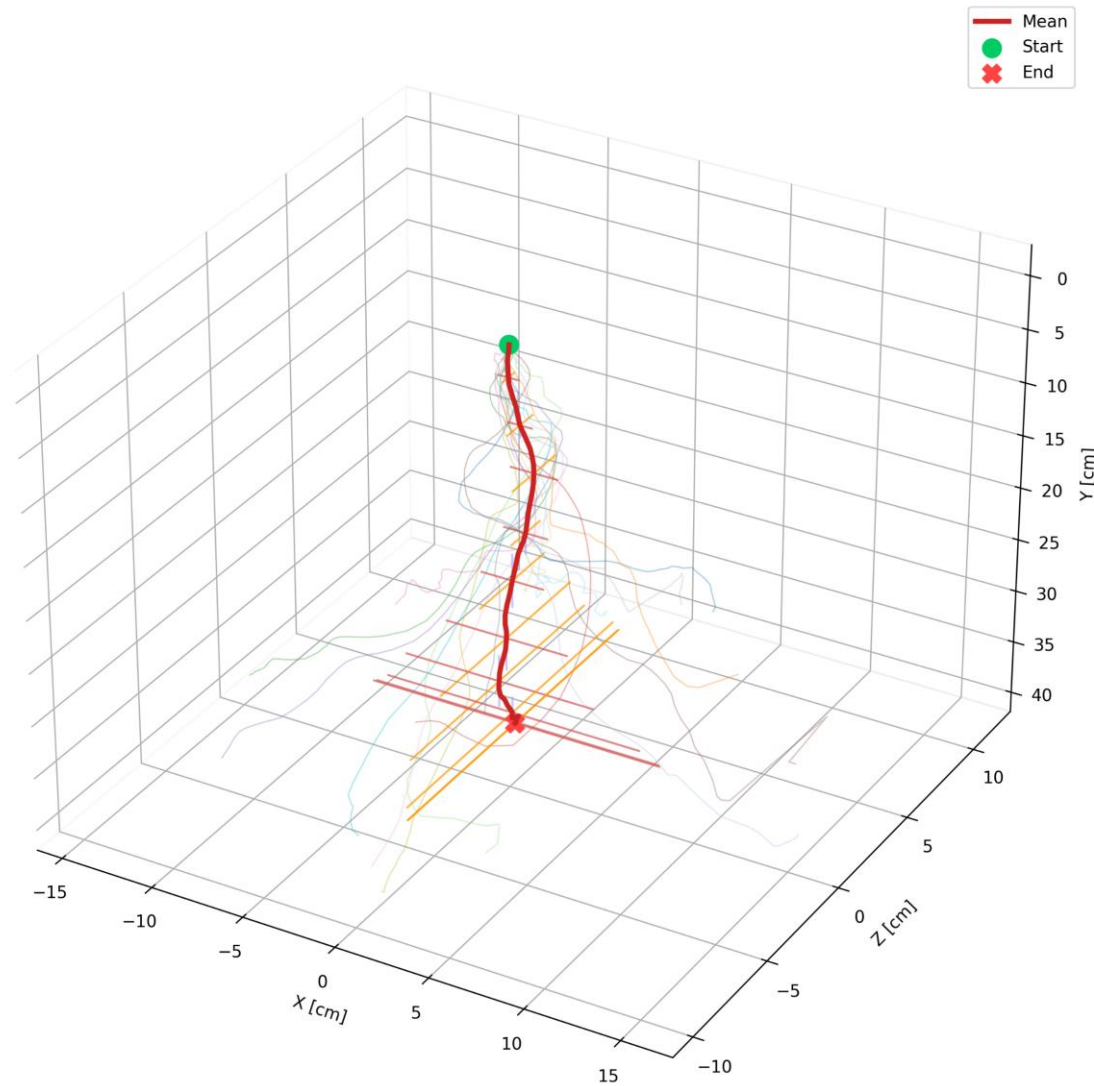


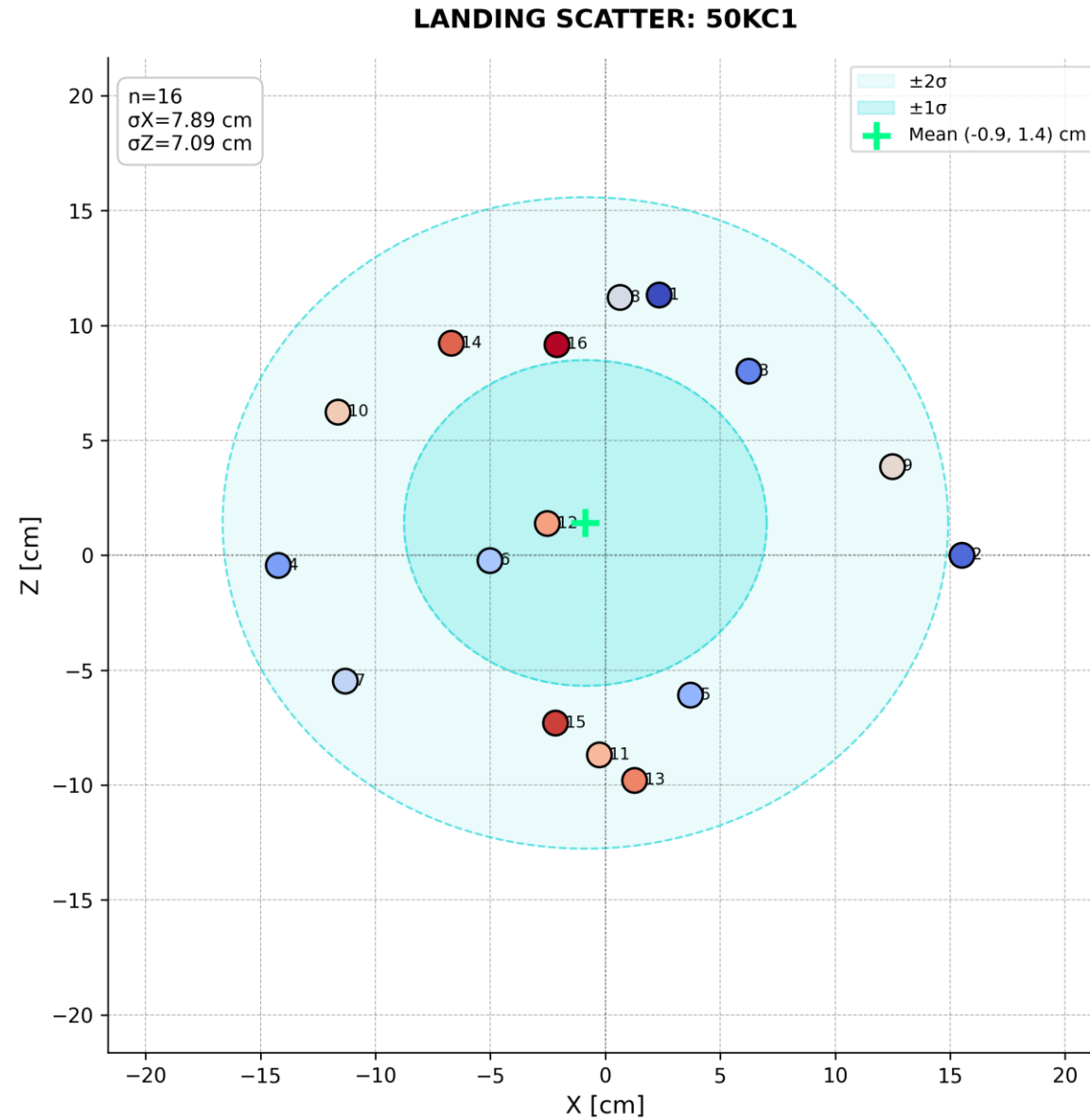
50 CZK

$m = 9.7 \text{ g}$, $d = 27.5 \text{ mm}$ $n = 16$



3D TRAJECTORIES: 50KC1 (n=16)

**50 CZK** $m = 9.7 \text{ g}$, $d = 27.5 \text{ mm}$ $n = 16$

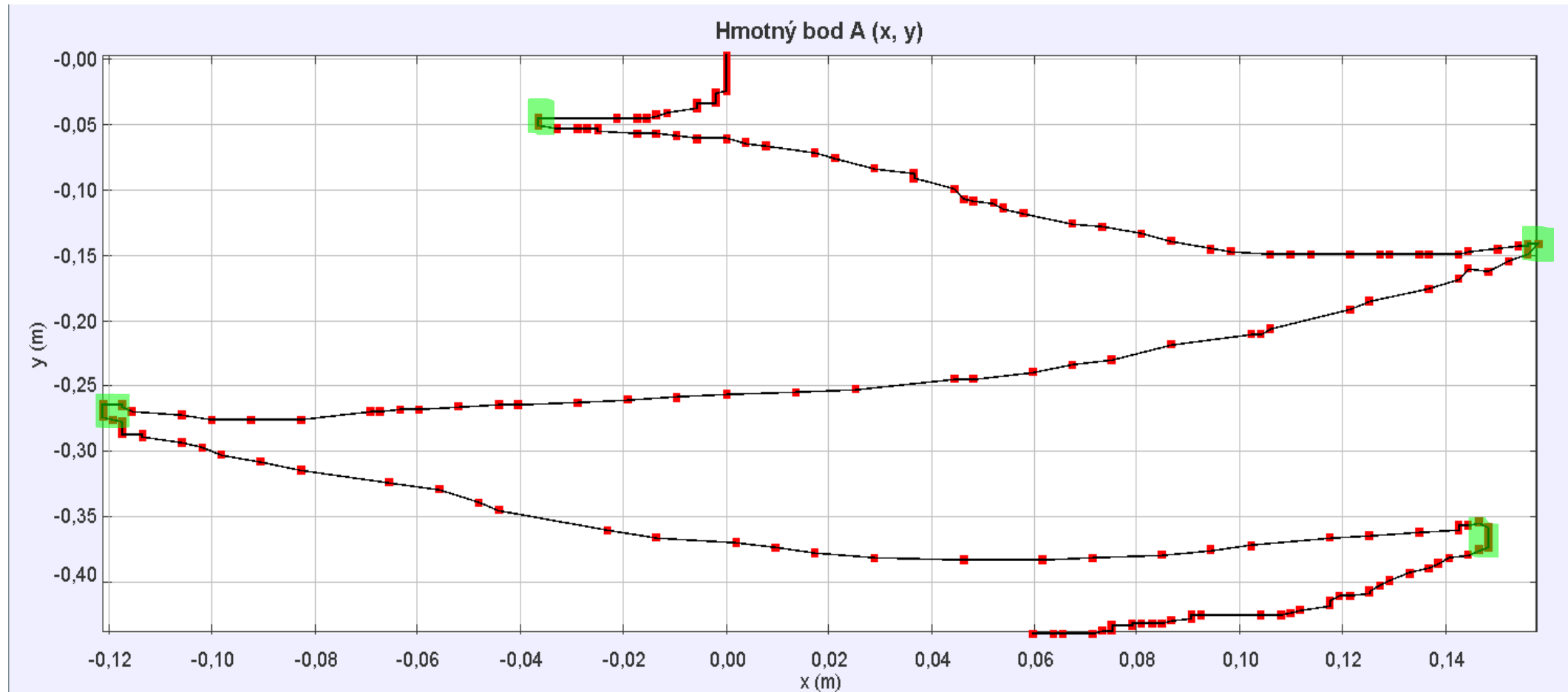


50 CZK

$m = 9.7$ g, $d = 27.5$ mm, $n = 16$

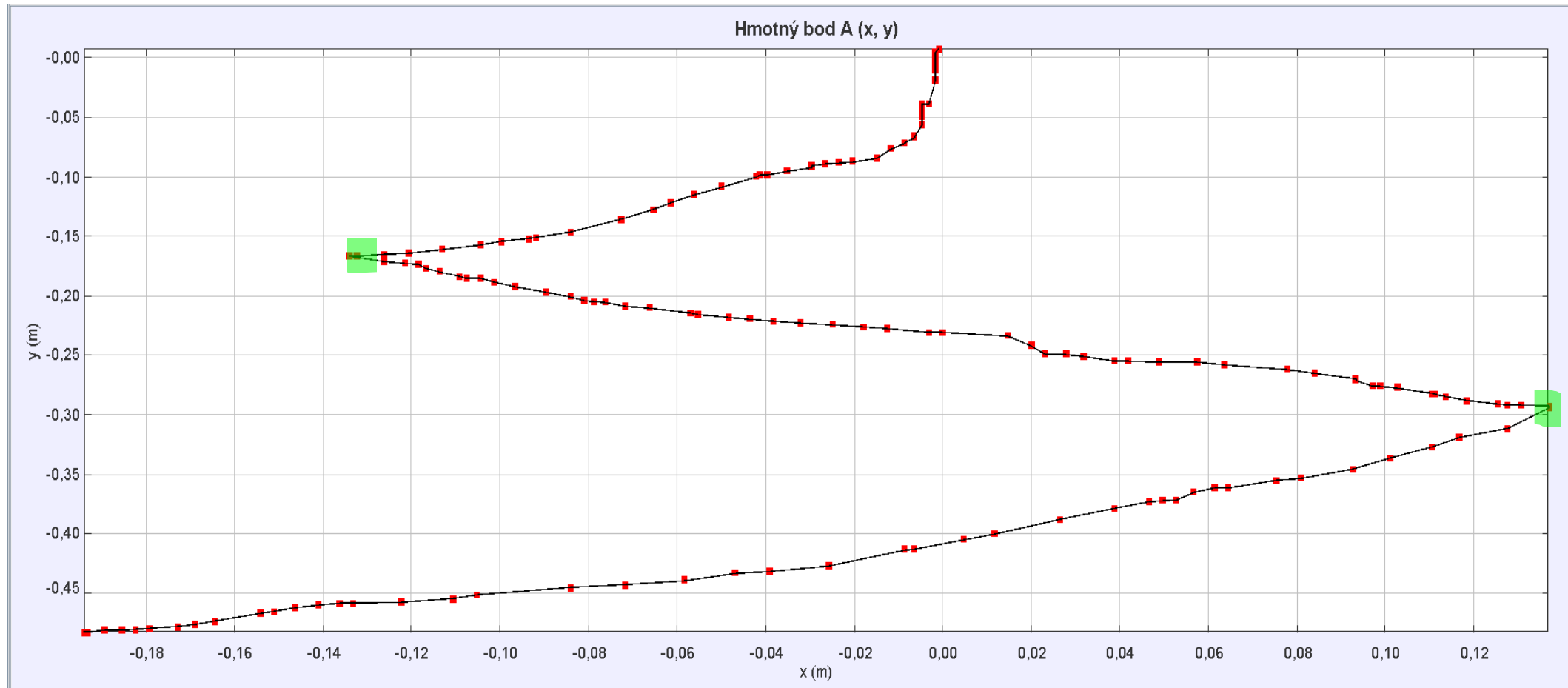
LIGHT WASHER WITH SMALL HOLE

4.7 g



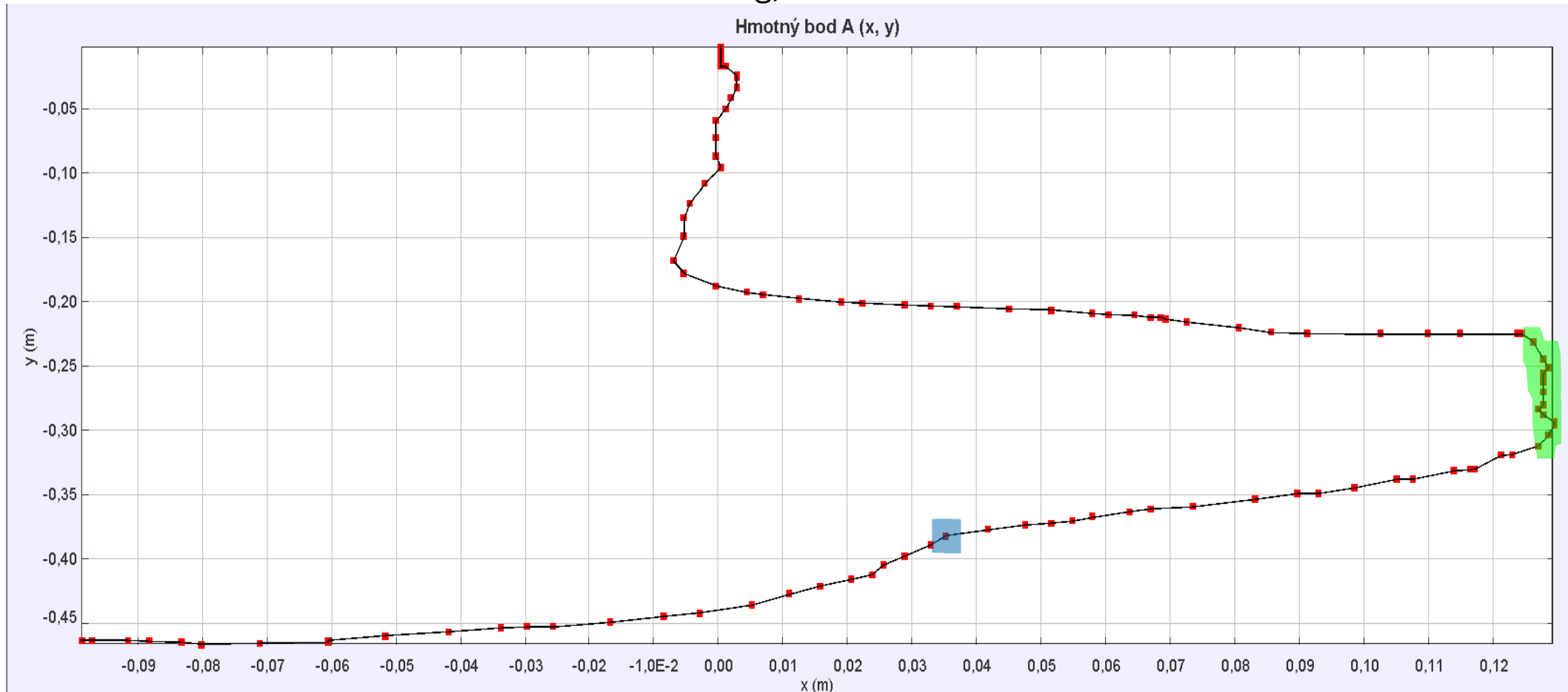
HEAVY WASHER WITH LARGE HOLE

11.9 g



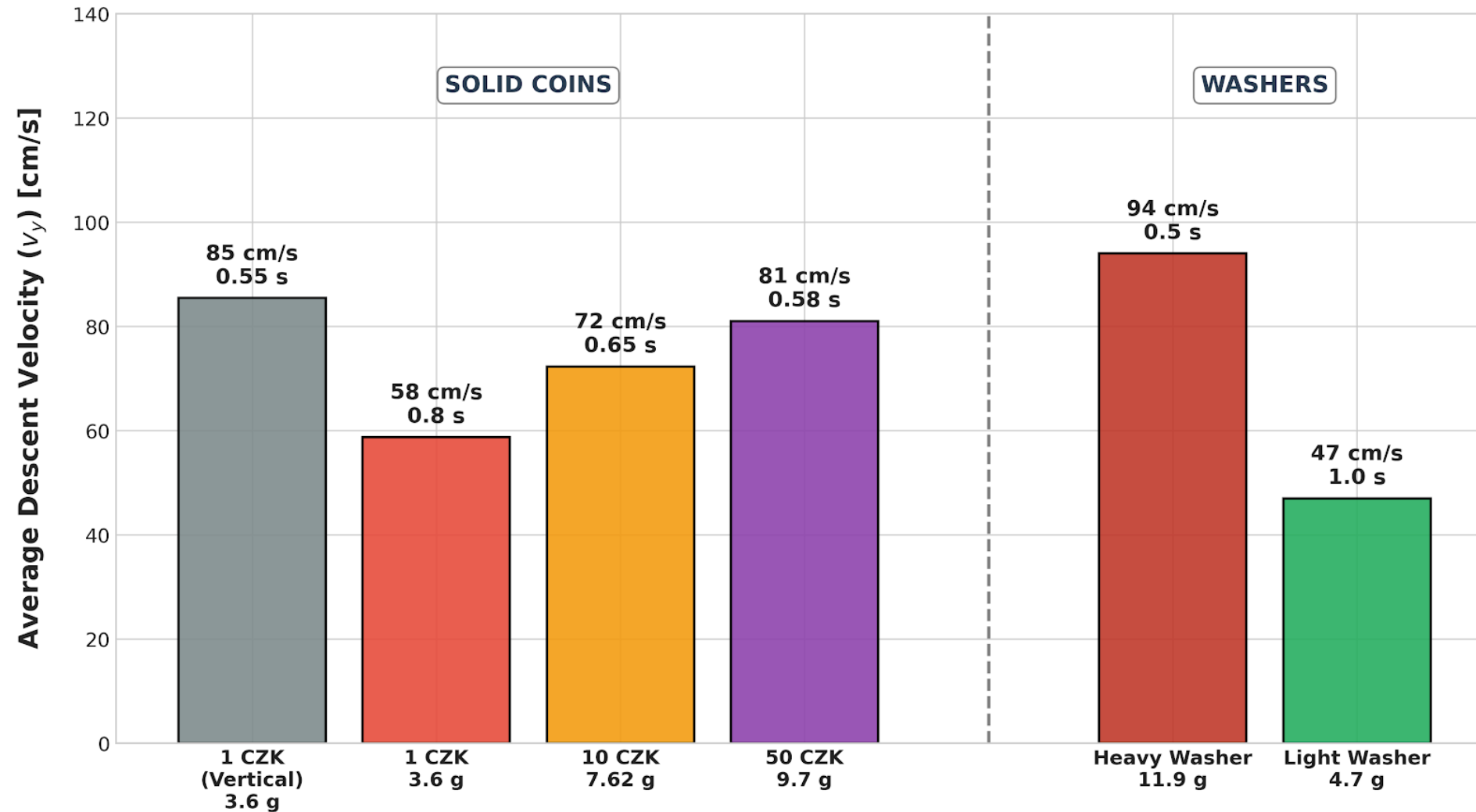
1 CZK (90° DROP)

$m = 3.6 \text{ g}$, $d = 20 \text{ mm}$





Final Comparison: Solid Coins (Left) vs. Washers (Right)





MASS VS. GEOMETRY

- Mass is not the solution
- Geometry is the key
- Light washer = stable glider
- > falls 50% slower than heavy washer

Sources:

- ▣ P. Martin. Falling leaves simulation (dspace.cvut.cz, 2022),
https://dspace.cvut.cz/bitstream/handle/10467/101330/F3-DP-2022-Pazout-MartinMaster_Thesis.pdf
- ▣ N. M. Kamaruddin. Dynamics and performance of flying discs (PhD thesis, Univ. of Manchester, 2011), <https://www.escholar.manchester.ac.uk/api/datastream?publicationPid=uk-acmanscw:132975&datastreamId=FULL-TEXT.PDF>
- ▣ U. Pesavento. Unsteady aerodynamics of falling plates (PhD thesis, Cornell Univ., 2006),
<https://ecommons.cornell.edu/items/06483654-e7b8-4900-8922-42bc3b6c7bfa>
- ▣ Andersen, A., Pesavento, U., & Wang, Z. J. (2005)
- ▣ Analysis of transitions between fluttering, tumbling and steady descent of falling cards
<https://www.cambridge.org/core/journals/journal-of-fluid-mechanics/article/abs/analysis-of-transitions-between-fluttering-tumbling-and-steady-descent-of-falling-cards/7EA576F55B2B0AEC088DCD62795C1372>